

Theory of Automata

Assignment # 2

Deadline: 31 October 2025 Instructions:

1. Attempt all questions.
2. Questions must be attempted on paper (A4 sheets or loose sheets) 3. Submit the assignment on time. Only physical submissions are accepted.

Part A – Kleene's Theorem: (2x3 + 2x3 = 12 marks)

Questions 1:

Kleene's Theorem Part 3 states that for every regular expression, there exists an equivalent finite automaton.

Kleene's Theorem Part 2 states that for every finite automaton, there exists an equivalent regular expression.

Task:

- a) Using Kleene's Theorem Part 3, construct an NFA for the regular expression $(a+b)^*ab$.
- b) Using Kleene's Theorem Part 2, derive the equivalent regular expression for the following finite automaton:

States = $\{q_0, q_1, q_2\}$

Alphabet = $\{a, b\}$

Start State = q_0

Final State = q_2

Transition Table:

$q_0 \xrightarrow{a} q_1$

$q_0 \xrightarrow{b} q_0$

$q_1 \xrightarrow{a} q_1$

$q_1 \xrightarrow{b} q_2$

$q_2 \xrightarrow{a} q_2$

$q_2 \xrightarrow{b} q_2$

Show all construction and derivation steps clearly.

Questions 2:

- a) Construct a regular expression corresponding to the finite automaton given below using Kleene's Theorem Part 2.

States = $\{S, A, B\}$

Alphabet = $\{0, 1\}$

Start State = S

Final State = B

Transitions:

S $\xrightarrow{0}$ S
S $\xrightarrow{1}$ A
A $\xrightarrow{0}$ B
A $\xrightarrow{1}$ S
B $\xrightarrow{0}$ A
B $\xrightarrow{1}$ B

b) Convert the regular expression $(0+1)^*11$ into an equivalent NFA using Kleene's Theorem Part 3.

Show all construction steps clearly.

Part B - Grammar, FA, PDA, CFG and Derivations: (20+9+9+4+9+10+10 = 71 marks)

Questions 1:

Design Regular Grammar and PDA for given Regular Language.

1. $01^*(0+1)^*$
2. $01(0+1)^*$
3. Letter followed by digit
4. Start and end with a
5. Generate all even integer upto 998
6. Alternate letters
7. a^{2n}
8. $(ab)^n$
9. $\{012, 120, 201\}$ generate these strings
10. Construct a grammar for the set of all strings over $\{0,1\}$ consisting of equal number of 0's and 1's.

Questions 2:

Draw the equivalent Finite Automaton (FA) for each grammar.

- a. S $\rightarrow aA \mid bB$
A $\rightarrow aA \mid bS \mid \epsilon$
B $\rightarrow bB \mid aS$
- b. S $\rightarrow 0S \mid 1A$
A $\rightarrow 0A \mid 1$
- c. S $\rightarrow 0A \mid 1B$
A $\rightarrow 1S \mid 0A$

B 0S | 1B

Questions 3:

For each FA (drawn or described), write the corresponding Right Linear Grammar or Context-Free Grammar.

- DFA that accepts strings ending with “01”.
- NFA for language $L = \{ w \mid w \text{ starts with } a \text{ and ends with } b \}$.
- NFA that accepts all strings containing substring “aba”.

Questions 4:

Convert each grammar as asked.

- a. Convert the following **LLG** → **RLG**

$S \rightarrow Aa \mid Ab$
 $A \rightarrow Ba \mid Bb \mid a$

- b. Convert the following **RLG** → **LLG**

$S \rightarrow aA \mid bB$
 $A \rightarrow a \mid aS \mid b$
 $B \rightarrow b \mid bS \mid a$

Questions 5:

For each grammar:

- Derive the given string using substitution (step-by-step).
- Show Leftmost and Rightmost derivations.
- Draw the Parse Tree.

1. $S \rightarrow aSb \mid ab$

String: **aabb**

2. $S \rightarrow aS \mid Sb \mid ab$

String: **aaabb**

3. $S \rightarrow 0S1 \mid 01$

String: **000111**

Questions 6:

Design Context Free Grammar and PDA for given Context Free Language.

- I. To design a context free grammar which can produce balance parenthesis consider in alphabet set we have $\{ (,) \}$
- II. Design a CFG and PDA for even and odd palindrom
a. Even palindrome 001 100
b. Odd palindrome 00100
- III. Design a CFG and PDA for language $\{01(1100)^n 110 (10)^n \mid n \geq 0\}$
- IV. Design a CFG and PDA for language $\{0^n 1^m \mid n \neq m\}$
- V. Design a CFG and PDA for language $\{ (0^n 1^n 2^m 3^m \mid n \geq 1, m \geq 1) \text{ or } (0^n 1^m 2^m 3^n \mid n \geq 1, m \geq 1) \}$
- VI. Design a CFG and PDA for language $\{ (0^i 1^j 2^k \mid k \leq i, k \leq j) \}$
- VII. Design a CFG and PDA for language $\{ (0^i 1^j 2^k \mid i=j, j=k) \}$
- VIII. Design a CFG and PDA for language $\{ (0^n 1^m 2^{2m} \mid n, m \geq 0) \}$
- IX. Design a CFG and PDA for language $\{ (0^n 1^m \mid n \leq m+3) \}$
- X. Design a CFG and PDA for language $\{ (0^n 1^m \mid 2^n \leq m \leq 3^n) \}$

Questions 7:

Scenario-Based Question:

You are developing a **syntax validation module** for a simple **programming language interpreter** that checks **nested structures** like parentheses, curly braces, and square brackets.

To model this system, you are required to **design a Pushdown Automaton (PDA)** that verifies the correctness of **balanced and properly nested symbols** in program code.

Description of the Problem:

A valid program string contains balanced pairs of $()$, $\{\}$, and $[]$ such that:

1. Every opening symbol has a corresponding closing symbol of the same type.
2. Symbols are **properly nested** — e.g., $\{ [()] \}$ is valid but $\{ () \}$ is not.
3. The input alphabet $\Sigma = \{ (,), \{, \}, [,] \}$.

Tasks:

- a) Define the **formal 7-tuple** $(Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ of the PDA that accepts valid nested symbol strings.
- b) Explain the **working principle** of your PDA (push and pop behavior).
- c) Draw the **state transition diagram** of your PDA.
- d) Trace the **step-by-step stack operations** for the input string $\{ [()] \}$.
- e) Write an **equivalent Context-Free Grammar (CFG)** that generates the same language accepted by your PDA.

Part C – Grammar Simplification: (1x5 = 5 marks)

Consider the following CFG for non-empty language:

S $\rightarrow aSa \mid bSb \mid a \mid b$

A $\rightarrow B \mid a$

B C | b

- Remove useless productions from the given CFG, if there are any.
- Check for the ambiguity in the given CFG for the string 'abba'.
- Convert the given CFG into PDA.
- Give the instantaneous description of resultant PDA.
- Trace the input string 'baab' on stack using the resultant PDA.

Part D – Minimization of DFA: (6+6= 12 marks)

Questions 1:

Given the following DFA:

States = {A, B, C, D, E}

Alphabet = {0, 1}

Start State = A

Final State(s) = {C, E}

Transition Table:

A --0--> B

A --1--> C

B --0--> A

B --1--> D

C --0--> E

C --1--> C

D --0--> E

D --1--> C

E --0--> E

E --1--> C

Task:

- Apply the DFA minimization algorithm to minimize the DFA.
- Show all intermediate steps clearly.
- Draw the minimized DFA.

Questions 2:

Consider a DFA defined as follows:

States = {q0, q1, q2, q3}

Alphabet = {a, b}

Start State = q0

Final State(s) = {q2, q3}

Transition Table:

q0 --a--> q1
q0 --b--> q0
q1 --a--> q2
q1 --b--> q1
q2 --a--> q3
q2 --b--> q2
q3 --a--> q3
q3 --b--> q3

Task:

- a) Identify all equivalent states.
- b) Construct the minimized DFA step by step.
- c) Write the new transition table of the minimized DFA.